Qn	Ans	Solution
1	A	units of $\Delta P = \dfrac{\text{kg m s}^{-2}}{\text{m}^2} = \text{kg m}^{-1}\ \text{s}^{-2}$ units of $\rho = \text{kg m}^{-3}$ units of $\left(\dfrac{\Delta P}{\rho}\right)^n = \left(\dfrac{\text{kg m}^{-1}\ \text{s}^{-2}}{\text{kg m}^{-3}}\right)^n = \left(\text{m}^2\ \text{s}^{-2}\right)^n$ units of $v = \text{m s}^{-1}$ For equation to be homogeneous, units of $\left(\dfrac{\Delta P}{\rho}\right)^n$ = units of v $\text{m}^{2n}\ \text{s}^{-2n} = \text{m s}^{-1}$ comparing indices of m: $2n = 1 \quad \Rightarrow \quad n = \dfrac{1}{2}$